Intelligent Patient Risk Assessment

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Abstract

This project implements a machine learning-based healthcare analytics system to assess patient health risks. Using the Diabetes Health Indicators dataset, the system predicts a continuous diabetes risk score and classifies the presence of diagnosed diabetes. The pipeline utilizes a cascaded approach: a Linear Regression model first estimates the risk score, which is then utilized as an engineered feature for a tuned Decision Tree Classifier. The system successfully processes structured clinical data to provide accurate risk predictions and insights.

1 Problem Statement

Intelligent Patient Risk Assessment is a critical step in predictive healthcare. The challenge addressed in this milestone is evaluating structured patient health data to accurately predict a continuous health risk score and classify the likelihood of a diabetes diagnosis. By relying strictly on traditional machine learning pipelines, this project establishes a robust, interpretable baseline for assessing patient deterioration probability before transitioning into agentic AI workflows.

2 Data Description

The project utilizes the **Diabetes Health Indicators Dataset** sourced from Kaggle.

- **Features:** The dataset contains structured clinical data, including patient vitals, lab values, and medical history.
- **Targets:** The system predicts two target variables: `diabetes_risk_score` (continuous) and `diagnosed_diabetes` (categorical).
- **Exclusions:** Non-clinical demographic variables such as ethnicity, education level, employment status, and income level were excluded to ensure the model focuses strictly on physiological and lifestyle health indicators.

3 Exploratory Data Analysis (EDA) Process

Before model training, a comprehensive EDA was conducted to understand the data distribution and identify potential anomalies:

- **Data Quality Checks:** The dataset was evaluated for unique values, missing values, and duplicated rows.
- **Numerical Features:** Boxplots were generated for numerical columns to identify outliers and understand the spread of clinical measurements.
- **Categorical Features:** Histograms, pie charts, and countplots were utilized to visualize distributions. Specifically, categorical features were

plotted against the `diagnosed_diabetes` target variable to observe distinct correlational trends between specific health indicators and diabetes presence.

4 Methodology

The system employs a cascaded machine learning architecture:

- **Data Preprocessing:** Categorical variables were transformed using `LabelEncoder`. The dataset was split into 80% training and 20% testing sets. Numerical features were standardized using `StandardScaler` to ensure uniform scale for linear modeling.
- **Risk Score Prediction (Linear Regression):** A baseline Linear Regression model was trained on the scaled features to predict the `diabetes_risk_score`.
- **Diagnosis Classification (Decision Tree):** The predicted risk scores were appended to the feature set as an engineered feature. A Decision Tree Classifier was then trained on this combined dataset to predict `diagnosed_diabetes`. A tree-based model was chosen for this step due to its high interpretability in clinical settings.

5 Optimization

Several optimization steps were taken to improve model performance and reliability:

- **Hyperparameter Tuning:** `GridSearchCV` was implemented to find the optimal parameters for the Decision Tree. The grid searched across `max_depth`, `min_samples_split`, `min_samples_leaf`, and `criterion`, using 5-fold cross-validation optimized for the `f1_weighted` metric.
- **Data Clipping:** To ensure clinical validity, the continuous predictions from the Linear Regression model were clipped to a strict boundary of 0 to 100 before being passed to the classifier.

6 Evaluation

The models were evaluated using standard regression and classification metrics.

Linear Regression Performance (Risk Score):

- **R-squared (R^2):** 0.9933
- **Mean Squared Error (MSE):** 0.5544
- **Mean Absolute Error (MAE):** 0.4391

Decision Tree Classifier Performance (Diabetes Diagnosis):

Metric	**Score**
Accuracy	0.9194
Precision	0.9980
Recall	0.8666
F1-Score	0.9276

Table 1: Performance metrics for the optimized Decision Tree Classifier.

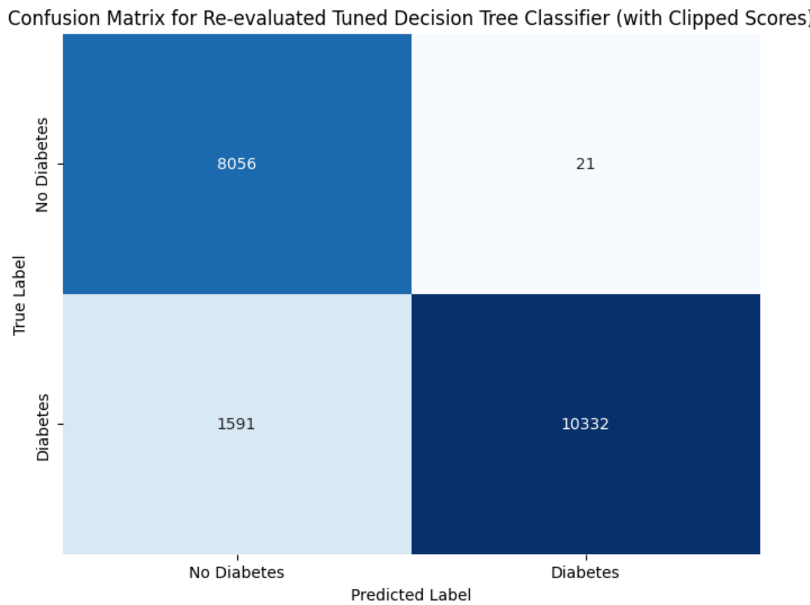

Figure 1: Confusion Matrix for the Re-evaluated Tuned Decision Tree Classifier

7 Team Contribution

- **Vikrant Yadav:** Led Data Preprocessing, Feature Engineering, and Exploratory Data Analysis (EDA).
- **Mayank Pillai:** Developed and evaluated the Linear Regression model for continuous risk scoring.
- **Prashant Raj:** Implemented the Decision Tree Classifier, executed GridSearchCV optimization, and generated evaluation metrics.
- **Asad Ali:** Managed LaTeX report documentation and will lead the Streamlit UI integration for the hosted demo.